EquationQuest AI | Brief Project Report

Course: Educational Technology & AI **Topic:** AI for Differentiated Learning **Math focus:** Linear Equations

1. Purpose and Design

EquationQuest AI is a responsive web-based learning application built with HTML, CSS, and JavaScript. It supports differentiated instruction by presenting the same core algebra objective—solving linear equations—through three progressively demanding forms. Learners select a level, solve generated equations, receive immediate answer feedback, and inspect targeted hints or worked steps.

2. How Differentiation Works

Foundation: One-step equations such as x + 5 = 9. The guidance names the inverse operation and explicitly models isolating x.
Standard: Two-step equations such as 3x - 4 = 11. Hints suggest the sequence of inverse operations while preserving student independence.
Advanced: Equations with variables on both sides, framed as matching two totals. Learners must collect like terms, balance both sides, and justify multiple steps.

3. AI Prompt Engineering

The optional secure AI tutor receives the selected level, current equation, request type (hint or full solution), and a mathematically verified reference solution. Its system instruction is to act as a patient algebra tutor, preserve the verified answer, and adjust scaffolding by level. For **Foundation**, it uses simple language and direct support; for **Standard**, concise algebraic reasoning; for **Advanced**, balancing and modeling explanations. Hint requests explicitly prohibit revealing the final answer.

4. Implementation and Verification

JavaScript generates equations from a chosen integer solution, making each displayed equation algebraically consistent. A deterministic checker validates student input, while progress counters show attempts, correct responses, and accuracy. Verified offline hints and steps work without an internet connection. The optional Cloudflare Worker connects to an AI API using a server-side secret; the browser never stores the key. Live AI responses require deployment and configuration of this backend.

5. Reflection

The project demonstrates differentiation through task complexity, scaffolding, and feedback while retaining one shared learning objective. AI can extend the prototype by adapting explanations dynamically; generated explanations should still be checked against the verified mathematical reference. Before final submission, the live website and AI endpoint must be deployed and tested.